

26-State HRes Spectral Ladder

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PAPER_952: 26-State HRes Spectral Ladder

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Date: 2026-04-12

Session: 214

Source: spectral_{ladder_26state}.py (SpectralLadder26State)

Calculator: SpectralLadder26StateCalc (CP4 #536)

CVW: v2.0.0 compliant

Abstract

We derive the 26-state HRes (Hydrogen Resonance) spectral ladder, the energy progression across quantum layers from proto-H ($n=1$) to proto-Fe ($n=26$). The energy levels $E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}$ form a geometrically growing ladder stabilizing the vacuum ratio $\rho_{\text{SCm}}/\rho_{\text{UA}} = 0.1$ and driving phonon resonance at every layer.

1. Energy Levels

$E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26} \cdot \frac{1}{\sqrt{S_{26}}}, \quad n = 1, \dots, 26$

where $E_0 = \hbar \omega_{\text{SCm}}$ and $\delta_n = (2\pi)^{n/6}$.

2. Element Mapping

n	Z_{id}	Element	Type
1	1	proto-H	Magnetic
14	14	proto-Si	Non-magnetic
26	26	proto-Fe	Magnetic

The ladder maps the complete magnetic/non-magnetic hierarchy from hydrogen through iron.

3. Source Data

- **File:** spectral_{ladder_26state}.py

- **Session:** 214

- **CP4 Class:** SpectralLadder26StateCalc (#536)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_953 — Ramanujan-Accelerated S_{26} Convergence
3. PAPER_956 — Spectral Ladder Phonon Mapping
4. PAPER_959 — 26D Ramanujan Summation

Cross-Links

Related Paper	Relationship
PAPER_953	Ramanujan acceleration of S_{26}
PAPER_956	Phonon mapping $E_n \rightarrow \omega_n \rightarrow Q_n$
PAPER_949	BCS gap uses ladder as phonon channels
PAPER_959	Full 26D Ramanujan summation engine

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2}\right)$$

where:

- L_{Edd} is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_{H} is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U,B,i}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i}} \right]$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^n)^n} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i}} \right]$$

This sum converges absolutely for $|S_{26}^{(3)}| < 1$ (satisfied by $|S_{26}^{(3)}| = 0.575$) and reduces to the classical Ramanujan $1/\pi$ series when $S_{26}^{(3)} \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4}	day ⁻¹ Magnetar spin-down
SS_q	—	0.57	BH dynamics
ω_{SCm}	—	$2\pi \times 1.25$ THz	Base energy E_0
E_0	$\hbar\omega_{\text{SCm}}$	8.27×10^{-22} J	Ladder base

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
26 energy levels	$E_n = E_0(2\pi)^{n/3} S_{26}$	Derived
Magnetic/non-magnetic hierarchy	Z mapping 1-26	Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:** Spectral Ladder (26-State HRes Phonon Hierarchy)**

§A.2 Lagrangian Density $\mathcal{L}_{\text{ladder}} = \sum_{n=1}^{26} \left[\frac{1}{2} \dot{q}_n^2 - \frac{1}{2} \omega_n^2 q_n^2 \right], \quad \omega_n = E_n/\hbar$

§A.3 Euler-Lagrange Equation of Motion $\boxed{\ddot{q}_n + \omega_n^2 q_n = 0, \quad n = 1, \dots, 26}$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow \omega_{\text{SCm}} \rightarrow$ spectral ladder $E_n \rightarrow$ 26-shell hierarchy $\rightarrow$ element mapping

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS $\rho_{\text{VDS}}(n) = \rho_{\text{SCm}} \cdot (2\pi)^{n/3}$ — density profile across ladder levels.

§B.2 DVP Prime sequence $\{2,3,5,7,11,13,17,19,23\}$ maps to magnetic shell closures at levels 2, 8, 14, 20, 26.

§B.3 BSH $\text{BSH}(n) = \tanh(n/26) \cdot S_{26}$ — saturation envelope over the 26 levels.

§B.4 Production-Scale Consistency

Metric	Value	Status
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VDS ratio	0.1	Confirmed
BSH layers	26	Confirmed
SSq	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
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PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_U B_i$ Jet Modulation
PAPER_1010	TON618 AGN $F_U B_i$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

19 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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